Prompted Recurrent Refinement for Solar Filament Instance Segmentation

A Solution to the Solar Filament Segmentation Challenge 2026

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ABSTRACT

This report describes our solution to the Solar Filament Segmentation Challenge 2026, a Kaggle competition on automatic segmentation of solar filaments in GONG H-α observations. We use a detect-then-refine pipeline whose intermediate outputs (proposals, keep/drop decisions, one mask per candidate) can be inspected separately. A frozen DINOv3 ViT-S+/16 with a 0.43 M-parameter head produces a high-recall filament probability map; its low-threshold connected components are filtered by a LightGBM classifier on 32 interpretable features; each kept component becomes the *prompt* of a recurrent refiner on DINOv3 ViT-B/16 that segments exactly one filament per prompt and predicts whether it exists. On the 707 labelled images, date-disjoint out-of-fold evaluation gives Panoptic Quality (PQ) 0.4405 (95% bootstrap interval 0.432–0.449), above the 0.34 of one expert annotator scored against another; without test-time augmentation it is 0.436, against 0.389 for the standard diffusion semantic baseline (DDP) trained on the same folds. The public leaderboard score is 0.39 (0.34 for that baseline). Code, weights and notebook: https://github.com/Cdshf01230/Solar-Filament-None00000.

1 INTRODUCTION

This report goes over the details of our submitted solution to the Solar Filament Segmentation Challenge 2026 [2]. The challenge, launched on July 10, 2026, is organized by the Earth-Space AI Research (ESAIR) Lab[1]. This challenge utilizes the MAGFiLO dataset for evaluation of the segmentation algorithms [1]. The input is a 2048^2 8-bit GONG [4] Hα image; the output is one binary mask per filament. Each of the 707 training images (2011–2022) has one to three annotation records (411/145/151 images): 1154 records and 8199 masks; 180 images form the test set. PQ [9] $= \sum_{TP} \mathrm{IoU}/(|TP| + \frac{1}{2}|FP| + \frac{1}{2}|FN|)$ with IoU > 0.5 is computed per (image, record). We report a strict variant with one-to-one Hungarian [10] matching, which duplicated masks cannot inflate; for our pipeline the organisers' all-pairs rule differs by at most 0.0001 PQ and one TP in every reported setting. Annotators disagree substantially: one annotator's masks scored against another's give PQ 0.340 (SQ 0.63) on the 296 multi-annotated images. We therefore never merge annotators: every training target and every score uses one record, on the full image. Because experts disagree on a filament's extent and identity, our longer-term goal is to help people find and correct annotations; we favour stages that can be inspected and used separately over combining many model families for a higher automatic score.

[1] www.esairlab.com/

Kaggle Competition, 2026, kaggle.com/competitions/filament-segmentation-2026
2026.

2 METHODOLOGY

Data split. The 707 labelled images are split once into five folds by observation date (seed 42; 141/141/142/141/142 images, 656 days), so near-identical images taken minutes apart never sit on both sides. The 180 test images are used only for the submission. Proposal heads and classifiers are trained on one fold each, and a labelled image of fold k uses the four that did not see it; refiner k is trained on the other four folds and predicts fold k; at test time all five models of each stage are averaged. Upstream models are reused to build downstream training inputs, so these are development out-of-fold (OOF) estimates, not fully nested cross-validation. Proposal threshold, filter threshold and last-epoch checkpoints are fixed; only post-processing was chosen on OOF.

Why detect-then-refine. Instances are not the obstacle in principle: splitting each record's ideal semantic mask into 8-connected components scores PQ 0.9989. With a predicted mask, however, a faint section splits one filament and a bridge of false pixels joins two, which is exactly what PQ punishes. On an earlier split (209 training images, 424 held-out) semantic heads of 1.6–2.1 M parameters on frozen DINOv3 [14] ConvNeXt-B features reached all-pairs PQ 0.30 (FPN [11]), 0.34 (U-Net [12]) and 0.37 (a small variant of the diffusion head DDP [6]). The best of them, DDP at its standard size (11.6 M parameters), trained for 20 epochs on the five folds above, still scores only 0.389 OOF, with about as many false positives as true positives (Section 3). These false positives are confident (median probability 0.95), so no threshold removes them. We therefore separate *where a filament may be* (cheap, high recall), *which candidates to keep* (explicit features) and *what exactly this filament is* (one instance per prompt by construction), see Fig. 1.

Pre-processing. The 8-bit image is read in grey, scaled to $[0, 1]$, repeated to three channels and normalised with ImageNet statistics for the backbones; no other filtering is applied.

Stage 1: proposal map. A frozen DINOv3 ViT-S+/16 provides block 3/6/9/12 features; a small FPN-like head with a half-resolution raw-image branch predicts per-pixel probability on nine 768^2 tiles blended with a floored Hann window. It is trained for four epochs with BCE (positive weight 4) + Tversky [13] (α=.2, β=.8) to favour recall. At $p \geq 0.05$ the OOF map has pixel recall 0.908 at precision 0.295; ViT-S+ was more precise than a frozen ConvNeXt-B at equal recall (19.3% vs 15.4% at 95% recall).

Stage 2: filter. The 16,721 OOF components of $p \geq 0.05$ (8-connected, $\geq$8 px) touch 99.3% of all filaments. A LightGBM [7]

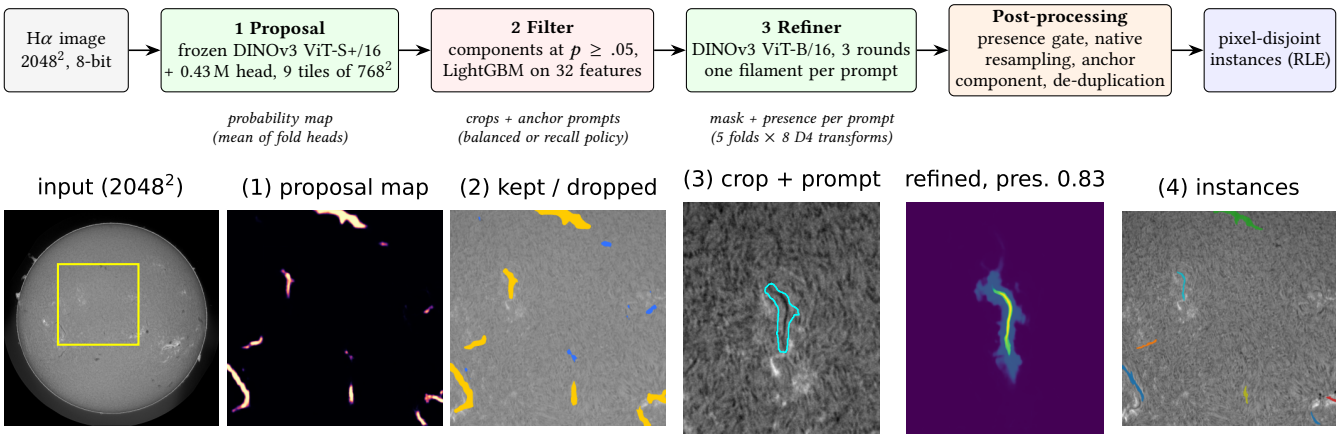


Figure 1: Top: the pipeline; every learned stage yields an output that can be inspected. Bottom: the stages on a test image (panels 1, 2 and 4 show the 768^2 window marked on the input). (1) Mean proposal map of the five fold heads; (2) components at $p \geq .05$ kept by the classifier (yellow) or dropped (blue); (3) one refiner crop with its prompt (cyan) and its round-3 probability averaged over 5 folds × 8 transforms; (4) final instances.

classifier (250 trees, 15 leaves) on 32 features (shape, probability statistics, darkness against Gaussian backgrounds, black-hat responses, thickness, elongation, context contrast) keeps components with mean score ≥ 0.5, and each kept component gives a crop (bounding box + 128 px). The default *balanced* policy keeps 5493 components, and 95.1% of the filaments intersect a retained crop. A released *recall-oriented* policy (positive class weight ×48) keeps 9411 (99.5%) for reviewing candidates; the refiner was trained on balanced crops and its PQ on recall crops has not been measured.

Stage 3: prompted recurrent refiner. The crop is letterboxed to 512^2 with five stem channels: image, proposal probability, anchor (the component), signed distance to it and current mask; letterbox validity is separate. A five-level convolutional stem is added at stride 16 to projected DINOv3 ViT-B/16 features (blocks 1–8 frozen, 9–12 fine-tuned with layer-wise LR decay 0.65 and drop path [5] 0.1), followed by four global self-attention blocks with a round embedding, so the two ends of a long filament can inform each other, and a U-Net decoder. Three recurrent rounds feed the previous mask back. A presence head predicts logit(ML score) plus a learned residual. The target is, per record, the filament overlapping the anchor most; prompts are the anchor (40%), a corrupted target (30%, eight corruption types), the model's own rollout (20%) or empty (10%). The loss sums BCE + Lovász hinge [3], presence BCE, an empty-crop term and Skeleton Recall [8] over all rounds, with D4, photometric and scale augmentation. Each fold model trains 30 epochs and the last checkpoint is used (34,820,802 trainable parameters).

Post-processing. Prompts with presence < 0.5 are dropped. The round-3 probability is resampled bilinearly to native size and thresholded at $p > 0.5$; the component overlapping the anchor most is kept (≥ 128 px). Candidates are accepted greedily by presence, skipping those with IoU > 0.5 to an accepted one and removing already-taken pixels (kept if ≥ 128 remain), so instances are pixel-disjoint. At test time the five fold refiners and eight dihedral transforms are averaged (40 passes per crop).

Table 1: OOF results (707 images, 1154 records, 8199 GT; strict matching). 95% intervals: bootstrap over images.

Setting	PQ	SQ	TP	FP	FN
threshold at 512, nearest upsampling	.4227	.669	4993	2621	3206
+ bilinear native resampling	.4363	.681	5065	2549	3134
+ D4 TTA (**submitted**)	**.4405**	.683	5100	2512	3099
95% bootstrap interval	[.432, .449]				
mean ± SD of the 5 folds	.441 ± .010 (range .430–.452)				
rerun of the released code (L4)	.4405	.683	5095	2498	3104
LOFO post-processing selection	.4395	.683			
DDP baseline, same folds, no TTA	.3886	.674	5557	5509	2642
annotator vs. annotator (all-pairs)	.340	.63			

3 EVALUATION

Strategy. All scores are OOF on the 707 labelled images, computed per original annotation record on the full image and counting filaments lost by any stage as false negatives; TP, FP, FN and the IoU sum are pooled over the 1154 records before PQ is computed. Uncertainty is given by a bootstrap over images (2000 resamples) and by the spread of the five folds.

Results. Table 1 gives the main scores. Resampling probabilities instead of masks improved 24/24 earlier model–fold pairs; test-time augmentation adds 0.004. Leave-one-fold-out (LOFO) selection among 11 stored post-processing settings gives 0.4395, a check of setting selection only. The organisers' self-evaluation code on our OOF predictions returns 0.440524, equal to our all-pairs scorer. One-change-at-a-time ablations on a single split showed the largest gains from D4 augmentation (+0.039), the ML-anchored presence (+0.009) and layer-wise LR decay with drop path (+0.007). The public leaderboard score is 0.39 (about half of the test images).

Dice, IoU and one-to-many / many-to-one. Fig. 3 gives the statistics of the organisers' self-evaluation notebook, computed per record on the OOF rerun of Section 4. The 6696 overlapping pairs have mean IoU 0.598 and mean Dice 0.730; the 5095 strict true positives have mean IoU 0.683 and mean Dice 0.808. Of the 8199

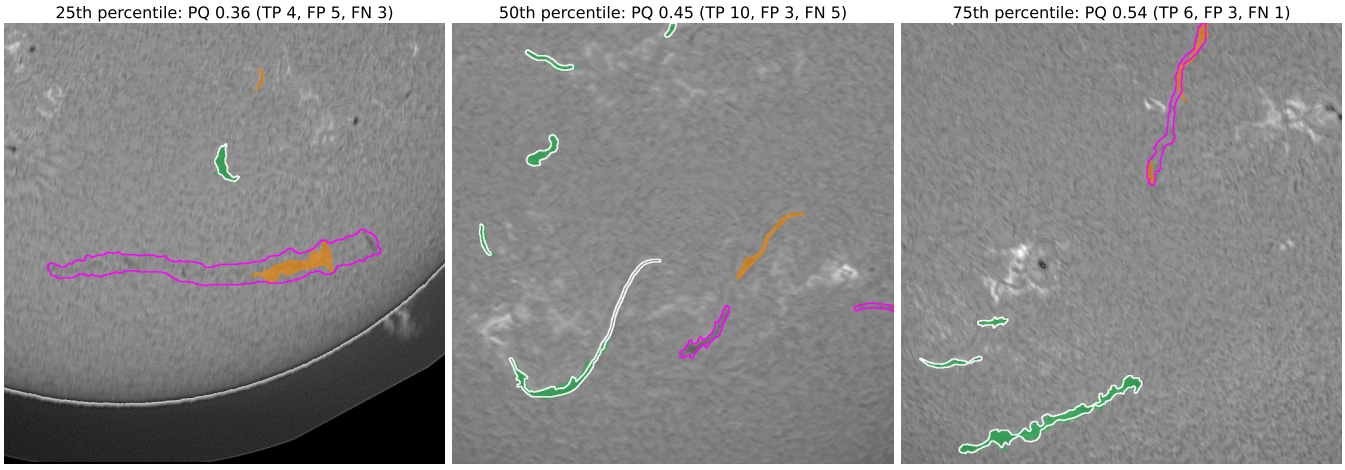


Figure 2: OOF predictions on the labelled images ranked at the 25th, 50th and 75th percentile of per-image PQ (among single-annotator images with at least six filaments; 896^2 windows). Green: prediction matched one-to-one with IoU > 0.5; orange: unmatched prediction; white outline: matched annotation; magenta outline: missed annotation. The typical error is a filament painted only in part (orange inside a magenta outline).

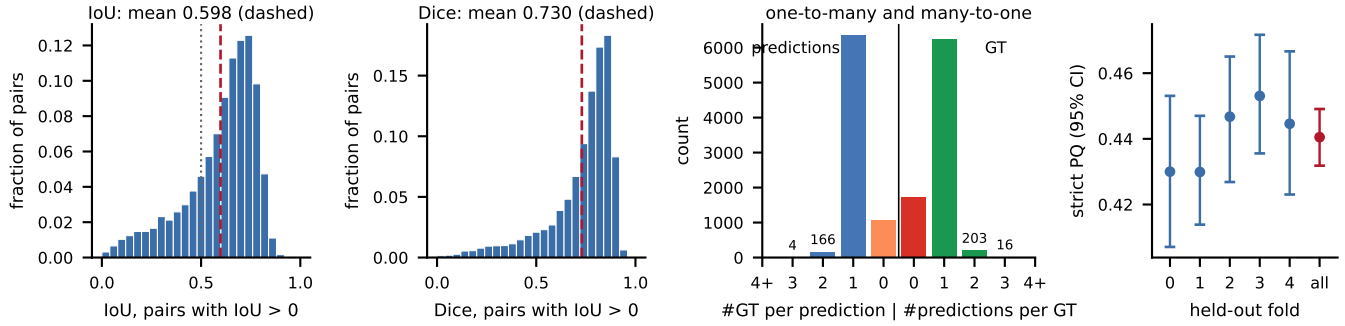


Figure 3: The organisers' rubric statistics on the OOF rerun (1154 records). Left to right: IoU and Dice of every (annotation, prediction) pair that overlaps (IoU > 0); number of annotations each prediction overlaps and number of predictions each annotation overlaps (degree ≥ 2: many-to-one and one-to-many); strict PQ per held-out fold and pooled, with 95% bootstrap intervals over images.

annotated filaments, 76% overlap exactly one prediction, 21% none and 219 (2.7%) two or more (one-to-many); 170 of 7593 (prediction, record) pairs (2.2%) overlap two or more filaments (many-to-one). The Dice of the union of the predicted masks against the union of a record's masks is 0.664 on average (0.690 pooled). Fig. 2 shows predictions on images of low, median and high PQ.

Efficiency. On one NVIDIA L4 the released `predict.py` processes the 180 test images in 84 min (28 s per image, models loaded once): stages 1–2 take 5.1 s per image (five proposal heads on one frozen backbone, components, features and LightGBM) and stage 3 takes 0.57 s per crop for each fold refiner with eight transforms (7.9 crops per image, five refiners). Stage 3 is 80% of the time; without TTA it is eight times cheaper and loses 0.004 PQ (Table 1). Trainable parameters are 0.43 M per proposal head, 250 trees per classifier and 34.8 M per refiner, on frozen DINOv3 ViT-S+/16 (28.7 M) and ViT-B/16 (85.7 M) backbones.

Semantic baseline. The standard DDP semantic head (11.6 M parameters, frozen ConvNeXt-B, threshold 0.5, 8-connected components), trained for 20 epochs on the same five folds with a cosine learning-rate decay that ends at 29% of its peak, scores 0.3886 OOF (choosing its epoch on the other four folds: 0.3924), against 0.4363 for our pipeline without TTA, with more than twice as many false positives (5509 vs 2549); its five-fold ensemble scores 0.34 on the public leaderboard against our 0.39.

Error analysis. The 3099 false negatives fall into disjoint classes: 1329 lie inside a crop but are touched by no prediction (514 only in crop padding), 558 are incomplete masks, 402 are lost before the refiner, 332 are too wide, 219 merged with a neighbour, 146 misaligned, 82 split and 31 taken by another match or by the presence gate. Across classes, 481 share an anchor with another filament and are never a crop target, since a prompt yields one instance. For incomplete masks 77% have a single anchor that already covers a

median 81% of the filament: the refiner sees the missing part but does not paint it. Removing every false positive on annotation background would raise PQ only to 0.473; background means unmatched to that record, not verified physical absence.

Strengths, weaknesses and improvements. Every stage has an output a person can inspect, each instance comes from an explicit prompt, and instances never overlap, so a filament cannot be counted twice. Its limits are the recall cost of filtering (a filament outside every crop cannot be recovered), incomplete masks and one output per prompt. Re-prompting on the part of an anchor the first mask does not explain finds up to 164 of the 506 never-targeted filaments, but at 14% precision, so the best zero-shot variant (0.4409) is within noise. A trained second output per prompt, mask completion and a refiner trained on recall-policy crops are the next steps. Support for annotation is an intended use that has not yet been validated in a user study.

4 CODE AND REPRODUCIBILITY

The repository contains the inference package (one module per stage, all constants in one file), a notebook that loads the weights from the repository, runs every stage with visualisations, reproduces the submission and evaluates OOF, pinned requirements, all weights (1.1 GB; refiner files store only the decoder and the four fine-tuned blocks, the rest verified bit-identical to the public DINOv3 weights) and the training code of every stage. On 12 November 2026 we downloaded the code and weights from the public Hugging Face copy (identical to GitHub), installed the pinned requirements on a Colab L4 and reran everything: the 180-image test CSV was byte-identical to the submitted one, and the OOF run on all 707 images gave PQ 0.440507 against 0.440463 recorded (555 images identical; the others differ slightly because stage-1 maps are recomputed instead of read from storage). Fresh training from scratch has not been repeated. Mirror: https://huggingface.co/None00000/Solar-Filament-None00000.

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